

Bell Nozzle Rocket Engine Equation Database

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$$C_nH_{2n+2} + \frac{3n+1}{2}O_2 \Rightarrow (n+1)H_2O + nCO_2 + energy \quad (1)$$

$$\dot{m} = \frac{m}{t} \quad (2)$$

$$\frac{1}{\gamma-1} = \sum \frac{mol\%}{\gamma_i-1} \quad (3)$$

$$\frac{P_x}{P_s} = \left(1 + \frac{\gamma-1}{2}Ma_x^2\right)^{-\frac{\gamma}{\gamma-1}} \quad (4)$$

$$\frac{T_x}{T_s} = \left(1 + \frac{\gamma-1}{2}Ma_x^2\right)^{-1} \quad (5)$$

$$\frac{\rho_x}{\rho_s} = \left(1 + \frac{\gamma-1}{2}Ma_x^2\right)^{-\frac{1}{\gamma-1}} \quad (6)$$

$$V_e = \sqrt{\frac{2\gamma}{\gamma-1}R_sT_c\left(1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma-1}{\gamma}}\right)} \quad (7)$$

$$R_s = \frac{P_x}{\rho_x T_x} \quad (8)$$

[1]

References

[1] test, "test." test, test.